and bi-directional capabilities.

To create high-quality, efficient standard IIoT infrastructure, these protocols are essential. Some well-known IIoT protocols are described below.

OPC UA. OPC Unified Architecture (OPC UA) is the next-generation standard from The OPC Foundation. Classic OPC, a well-known standard for industry, provides a standard interface to communicate with PLCs. It is a client-server protocol, where the client can connect, browse, read and write to industrial equipment.

Highly secure, it is a good solution for tying PLC and sensor data into existing industrial applications like SCADA and manufacturing execution systems, where OPC and OPC UA connectivity are already available.

MQTT. Message Queue Telemetry Transport (MQTT) is a lightweight messaging protocol designed for volatile communication networks like cellular or satellite. It does not require high bandwidth or high-availability network connections. MQTT is preferred by hardware manufacturers due to its efficiency, security and scalability.

CoAP. Constrained Application Protocol (CoAP) is a specialised Web transfer protocol for use with constrained nodes and constrained networks in the IoT. It is designed for machine-to-machine (M2M) applications, such as smart energy and building automation.

AMQP. Advanced Messaging Queuing Protocol (AMQP) is an open standard for passing business messages between applications or organisations. It connects systems, feeds business processes with the information these need and reliably transmits onwards the instructions to achieve goals.

ZeroMQ. This is a high-performance messaging library for communication between applications or processes regardless of hardware or software. It can be an ideal IIoT protocol, provided the applications do not require high-performance or real-time operation.

STOMP. Streaming Text Oriented Messaging Protocol (STOMP) is similar to HTTP and is designed for interoperability of brokers.

A new lease of life to industries

The global IIoT market is expected to reach US$ 933.62 billion by 2025, according to a report by Grand View Research. Growing demand and adoption of cloud computing coupled with scalability of IPV6-3.4 × 10^{38} IP address are said to drive the market.

The global IIoT market exceeded US$ 100 billion in 2016 and is expected to grow at a CAGR of over 25 per cent from 2017 to 2025. The Asia-Pacific market is anticipated to outgrow the North American market and emerge as the highest revenue-generating region, with China spearheading the growth by the end of 2025.

In 2015, the Chinese government launched Made in China 2025 strategy.

> GE predicts that US$ 1.3 trillion can be captured in the electricity value chain from 2016 to 2025 globally by the IIoT

This project aims to increase China's manufacturing innovation. The manufacturing sector of China has received in excess of US$ 1.5 billion with an objective to turn China into a manufacturing superhouse. As a result, the global IIoT industry is set to benefit from this.

The IIoT value chain across India

It might take a few years for the IIoT to be fine-tuned for wider industrial use. However, it can revolutionise business models by bringing smart manufacturing into play.

The manufacturing industry in India has shown phenomenal growth in the last decade. But India's ranking as per Global Manufacturing Competitiveness Index has dropped from number 2 in 2014 to number 11 in 2016. A key reason for this is the lack of focus towards building a skilled workforce and the inability to establish modern technologies.

Automation in manufacturing companies in India is done through